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UNIVERSITI
TEKNOLOGI
MARA

ICT602
MOBILE TECHNOLOGY AND DEVELOPMENT

CHAPTER 4 – MOBILE WEB APPLICATION

Topic Outline

1. Mobile Web Standards and W3C
2. Wireless Markup Language
3. XHTML Mobile Profile
4. HTML 5
5. Mobile Web vs Desktop Web
6. Mobile Web Era
7. Mobile Browsing Experience

MOBILE WEB STANDARDS

Mobile Web

- The widespread deployment of **Web-enabled mobile devices** such as phones make them a good choice for content developers.
- A number of the barriers that mobile users face are similar to those experienced by people with disabilities.
- Aim at developing **web sites that are accessible** for both disabled people and for **mobile devices**

Mobile Web

- Need of **Internet everywhere**
- Need of **same services** on every devices
- Need guideline to help developers
- Need guideline to improve the **interoperability, usability, and accessibility** of mobile web usage.

Mobile Web: Solution.

- A Standard
- To support **any mobile devices** using **Web Standards**
- **One Source Multi Use**

Mobile Web Standards

- Fundamentally, there is **one Web**.
- Its content is standardized markup, styles, scripts, and multimedia viewable using web browsers.
- A standards-based approach to Mobile Web development ensures **compliance and usability across mobile browsers & platforms.**
- Knowing all the rules & knowing when to ignore the rules is necessary for success on the Mobile Web.

WORLD WIDE WEB CONSORTIUM ROLES

Role of W3C

- World Wide Web Consortium (W3C) is the **main international standards organization** for the World Wide Web (WWW)
- W3C role includes ensuring the Web be available on as many kind of devices as possible, supporting different platform as widely as possible.
- With the explosion of powerful internet capable mobile devices in the past few years, W3C role to enable Web for those devices has become increasingly important.

Role of W3C

- W3C accompanies this growth with its ongoing work in the following areas:
  - Mobile Web Standards and API specification
  - W3C Widget
  - **Mobile Web Best Practices**
  - Guidelines on designing websites that are both accessible and Mobile Friendly

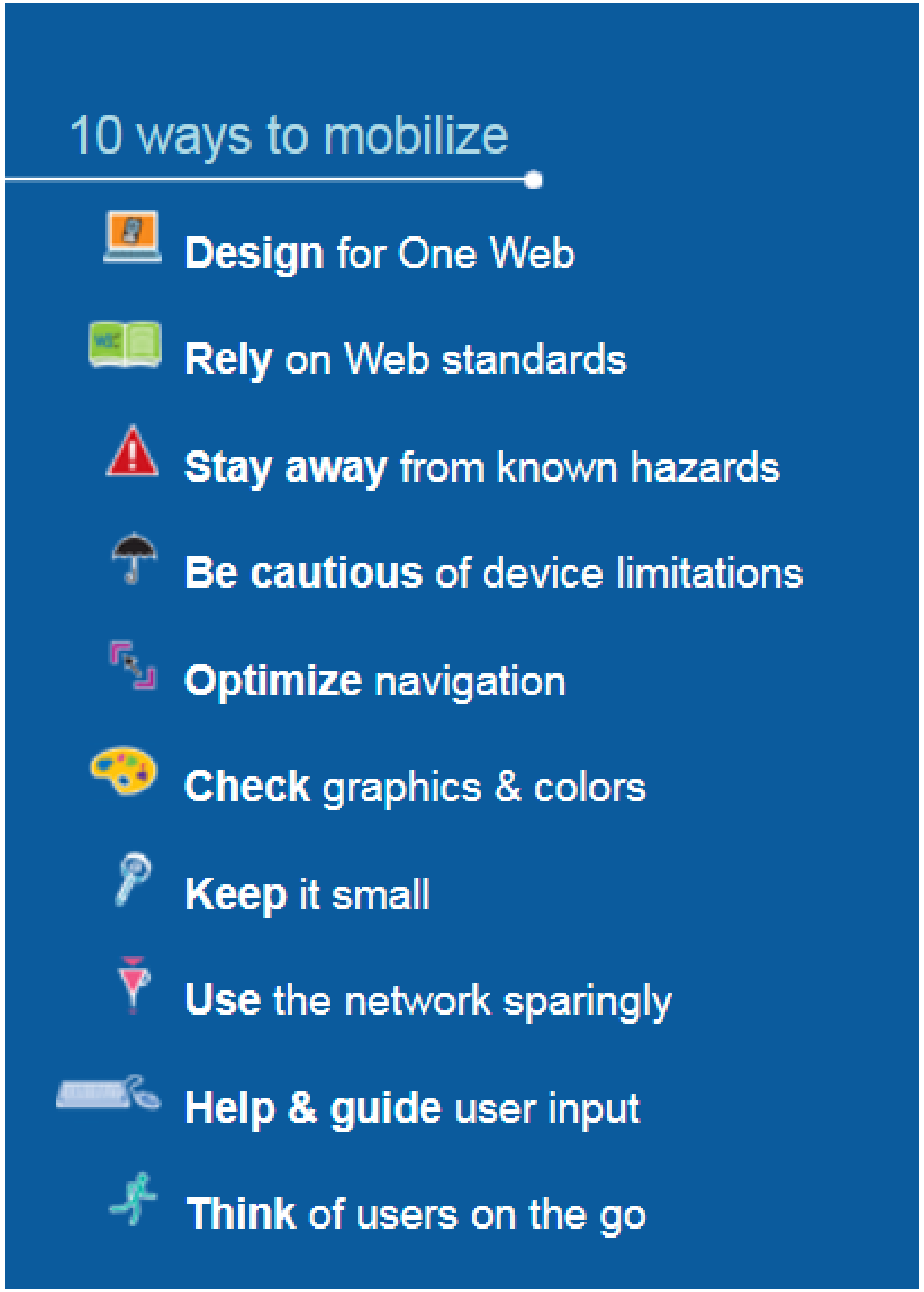
- https://www.w3.org/2012/05/mobile-web-app-state/
- https://www.w3.org/standards/techs/mobileapp#w3c_all

Mobile Web Best Practices

- "Mobile Web Best Practices 1.0" is a W3C Web Standard that **helps people design and deliver content that works well on mobile devices**.
- The principal objective is to **improve the user experience** of the Web when accessed from such devices.

Mobile Web Best Practices

- Set of recommendations finalized to improve the user experience
- Give advice about navigation and links, design, layout, content, technical issues and user input
- Not concentrated just on **technical** problems, but also **non-technical** like the delivered content



Mobile Web Standards

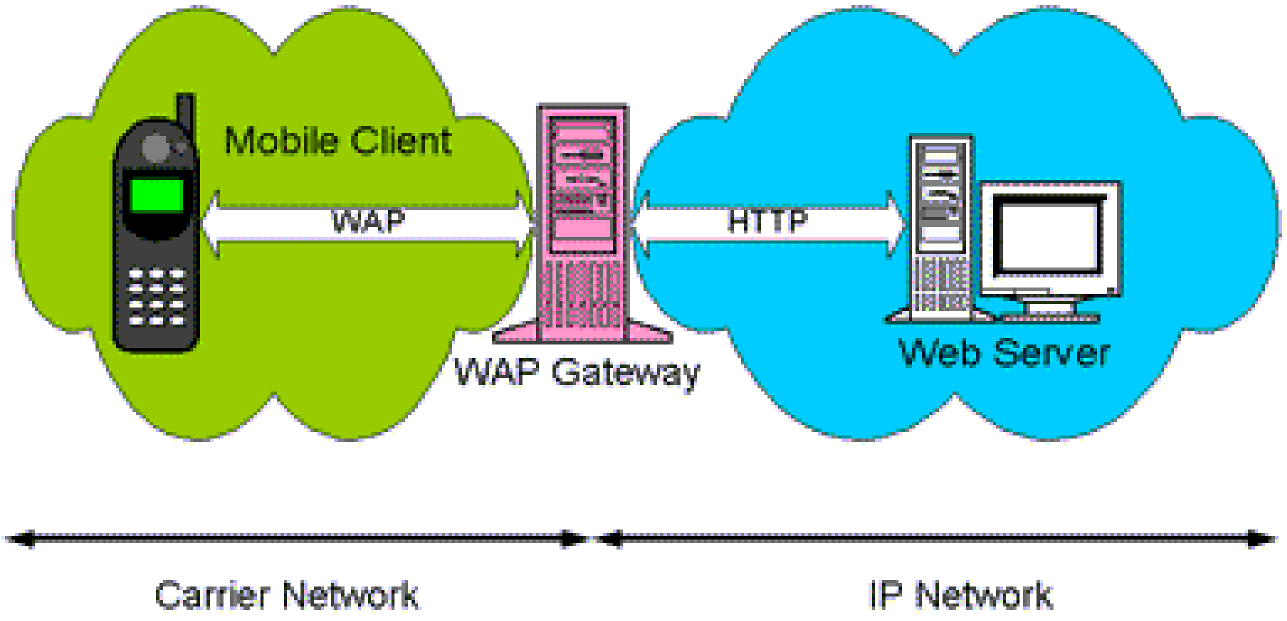
- There are other types of Mobile Web Standards including those that are proposed by W3C.

  - Wireless Mark Up Language (WML)
  - XHTML Mobile Profile (XHTML-MP)
  - HTML 5

WIRELESS MARKUP LANGUAGES

Wireless Markup Language (WML)

- WML is an **early standard** for **displaying mobile web** and implemented on top of Wireless Application Protocol (WAP)
  - WAP is a technical standard for accessing information over a mobile wireless network. A WAP browser is a web browser for mobile devices such as mobile phones that uses the protocol.



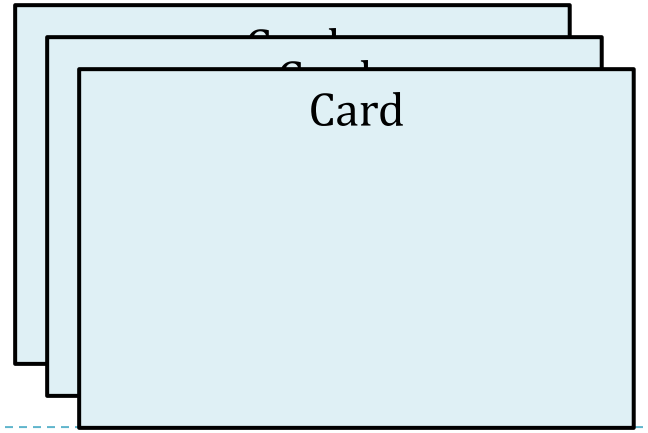
Wireless Markup Language (WML)

- WML is released by the WAP Forum and is based on **XML**
- The first Mobile phone model that are capable of displaying Mobile Web using WML is the Nokia 7110

Wireless Markup Language (WML)

WML **Structures**

1. Each WML document is made up of multiple **cards**
2. A Card represent a single screen
3. Cards can be grouped together into a deck
   - A deck is the smallest unit of WML a web server can send to a WAP browser



Wireless Markup Language (WML)

WML Basic **Features**

1. **Text and Images** can be presented to a user
2. The exact UI presentation **depends on the running device** (devices represent UI differently)
3. Offers **history mechanism** with navigation through the browsing history, hyperlink and other intercard navigation elements
4. WML allows saving the state between different decks **without server interaction**

Wireless Markup Language (WML)

Advantages

1. **Compact and small** in size
2. Can be compressed using binary encoding to further **save bandwidth** in mobile device
3. Saves state between cards within the deck without server intervention
4. UI element suitable for small-screen

Disadvantages

1. Limited UI Screen elements
2. Only supports **monochrome** images (WBMP)
3. UI Screen implementation varies with devices
4. Not suitable for displaying rich multimedia content

WML Example

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//OPENWAVE.COM//DTD WML 1.3//EN"
"http://www.openwave.com/dtd/wml13.dtd">

<!-- Canadian Provinces and Territories Main Screen -->
<wml>
<card id="main" title="Provinces and Territories" -->
    <p>
        <a href="#cardon">Ontario</a>
    </p>
    <p>Quebec</p>
    <p>British Columbia</p>
    <p>Nova Scotia</p>
    <p>Manitoba</p>
    <p>Alberta</p>
    <p>New Brunswick</p>
    <p>Prince Edward Island</p>
    <p>Newfoundland and Labrador</p>
    <p>Saskatchewan</p>
    <p>Northwest Territories</p>
    <p>Yukon</p>
    <p>Nunavut</p>
</card>
</wml>
```

XHTML MOBILE PROFILE

XHTML Mobile Profile

- eXtensible Hypertext Markup Language – Mobile Profile (XHTML-MP) is a specialization of XHTML designed to include features useful to mobile devices
- XHTML-MP 1.0 was defined by the (Open Mobile Alliance, OMA) an is an **extension** of the original W3C-inspired XHTML Basic 1.0
- Unlike WML, Mobile Web page developed using XHTML MP is **viewable in other HTML compatible browsers** (Google Chrome, Internet Explorer, etc)

XHTML Mobile Profile

Advantages

1. An extension to W3C developed XHTML Basic
2. Supported widely across different device platform
3. Can be displayed with web browsers compatible with XHTML / HTML
4. Supports Cascading Style sheet styling

Disadvantages

1. Developers may have to rely on ready-made templates
2. XTHML is only an extension, instead of designed from ground up for mobile devices

XHTML Mobile Profile Basic

```
<?xml version="1.0"?>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">

<!-- XHTML 1.0 Strict file created by the Openwave SDK -->

<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en">
    <head><title>XHTML Basic Example</title></head>
    <body>
        <p>Welcome to my first XHTML Basic Example</p>
    </body>
</html>
```

HYPER TEXT MARKUP LANGUAGE 5

Hypertext Markup Language 5 (HTML5)

- HTML 5 is the **current standard** released by W3C for developing Mobile web pages and Mobile Web Application
- HTML 5 consists of a set of Mark-up Languages derived from HTML 4.01 plus a set of new Modular Application Programming Interface (API)

```
<article>   <figcaption> <progress>
<aside>     <footer>     <section>
<audio>     <header>     <source>
<canvas>    <hgroup>     <svg>
<datalist>  <mark>       <time>
<figure>    <nav>        <video>
```

HTML

Goals of HTML5

- Support all existing web pages. With HTML5, there is **no requirement to go back and revise older websites**.
- **Reduce the need for external plugins and scripts to show** website content.
- Improve the semantic definition (i.e. meaning and purpose) of page elements.
- Make the rendering of web **content universal and independent** of the device being used.
- Handle web documents errors in a better and more consistent fashion.

Mobile HTML5

- **An HTML5 mobile app is a Web application developed with that version of the Web content standard and designed for smartphones, tablets and other handheld devices.**
- almost all current mobile devices support HTML5, which makes developing applications for multiple mobile platforms much simpler, as the code need be only written once

Hypertext Markup Language 5 (HTML5)

Advantages

1. HTML5 is a well-defined W3C standards and is supported by almost all modern devices
2. HTML5 API allows a certain degree of access to **Native Mobile features**
3. Support Mobile Web Application Templates and Framework

Disadvantages

1. Slightly high learning curve when compared to XHTML-Mobile Profile
2. API may not supported on some devices, or are implemented differently

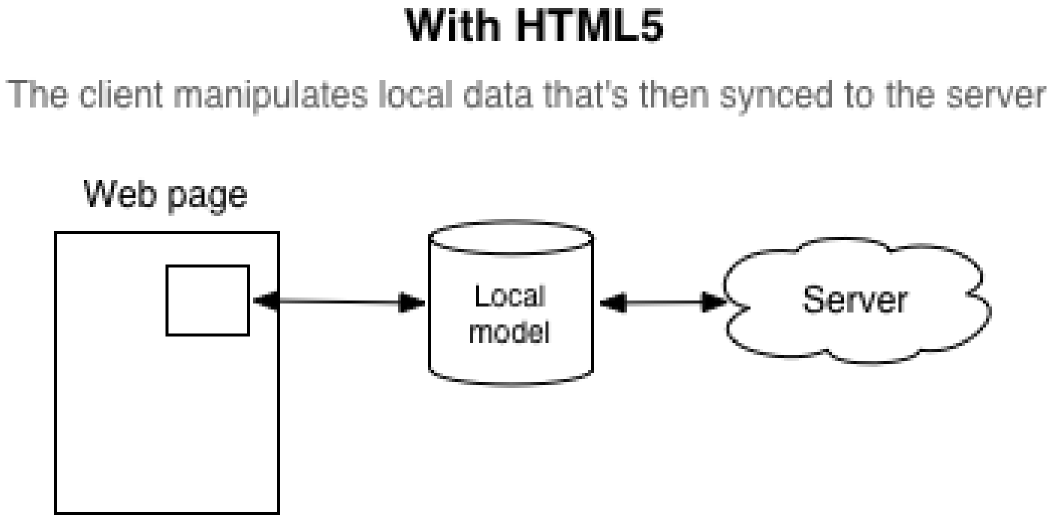
HTML5 APIs

- Among the API supported by HTML5 useful for building Mobile Web Application / Web are:

1. Web Storage
2. Geolocation
3. Viewport

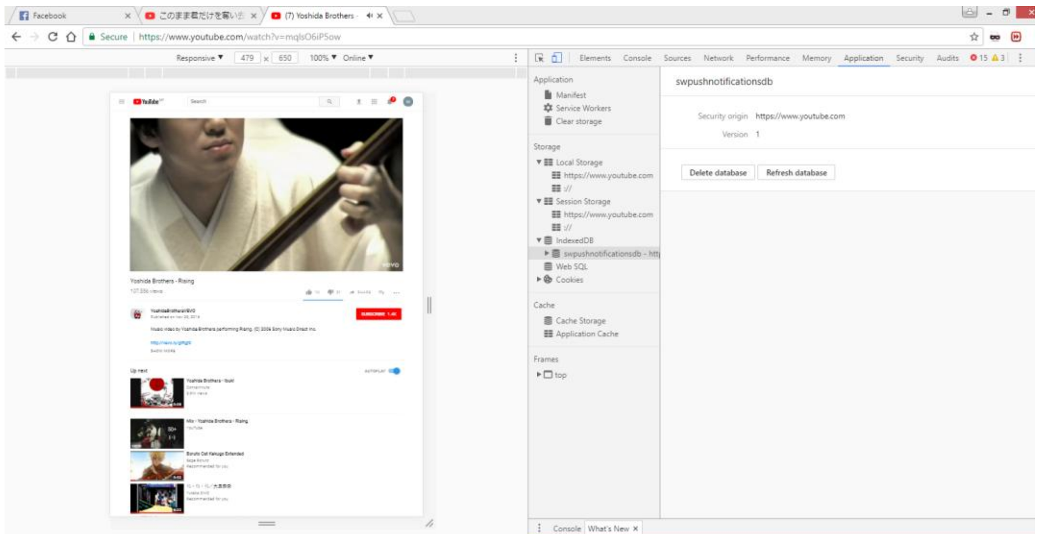
Web Storage

- HTML5 changes that, as there are now several technologies allowing the app to save data on the client device.
- It might also be sync'd back to the server, or it might only ever stay on the client.
- A simple client side database that allows the users to persist data in the form of key/value pairs



Web Storage

- Types of Web Storage
  - **Local storage:** Stores data with no expiration date. The data will be available even when the browser/ browsing tab is closed or reopened.
  - **Session storage:** Stores data for one session. Data persisted will be cleared as soon as the user closes the browse
- With local storage, web applications can store data locally within the user's browser.

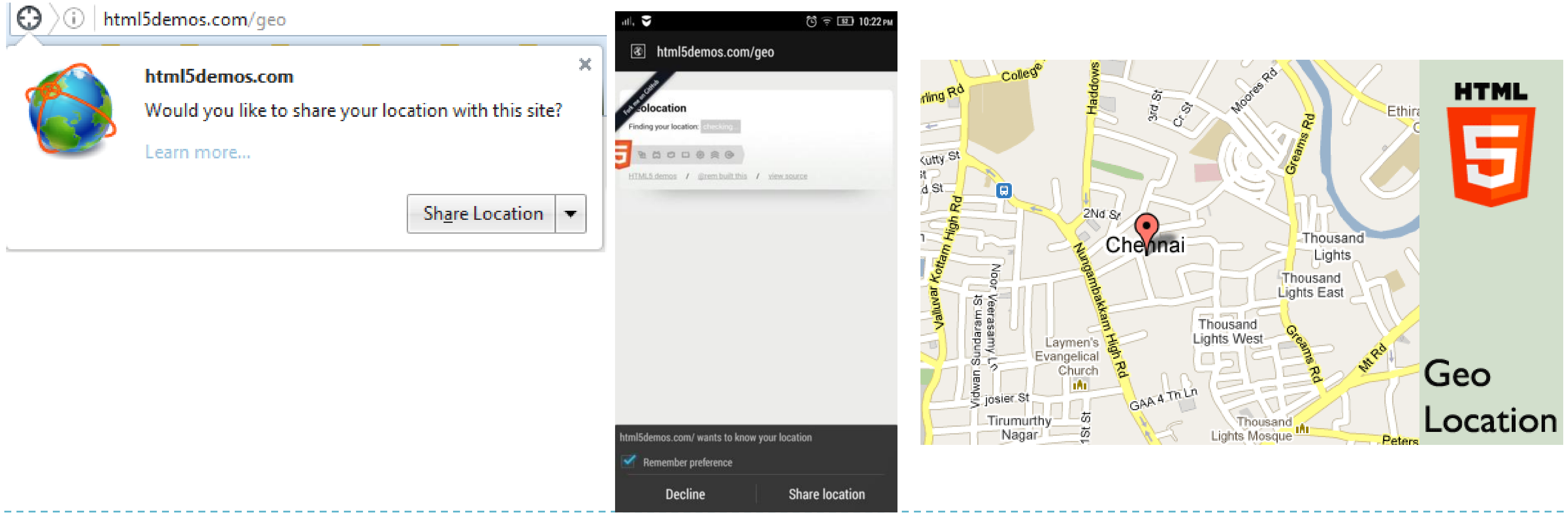


Web Storage

- **Advantages**
  - You can make your app work when the user is **offline**
  - You can show a large corpus of data as soon as the user clicks on to your site, instead of waiting for it to download again
  - **No server infrastructure** required
- **Disadvantages**
  - Can only store **strings**.
  - Data is more **vulnerable**
  - The user **can't access it from multiple clients**

HTML5 Geolocation API

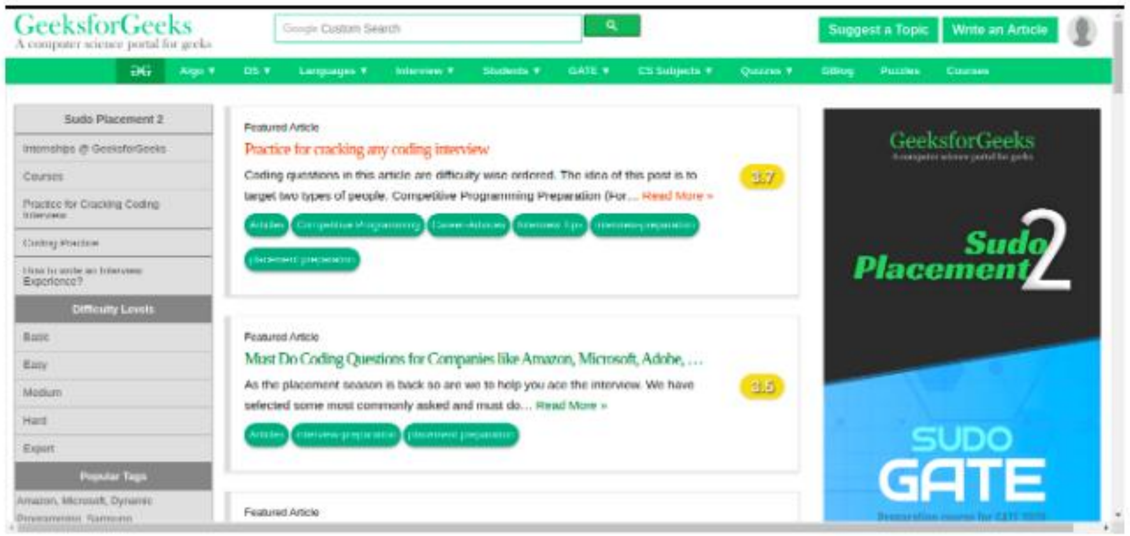
- The HTML Geolocation API is used to get the geographical position of a user.
- Geolocation is much more accurate for devices with **GPS**
- http://www.w3schools.com/html/html5_geolocation.asp



HTML5 Viewports

- The viewport is the user's visible area of a web page.
- The viewport varies with the device and will be smaller on a mobile phone than on a computer screen.
- HTML5 introduced a method to let web designers take control over the viewport, through the <meta> tag.

Example: The laptop screen is wider without the viewport meta tag.



Example: The screen is responsive with the viewport meta tag .On smartphones the page is adjusted due to narrower screen

Other APIs Useful for Mobile

- Vibration
- Device Orientation
- Battery Status
- Notifications
- Form Virtual Keyboards
- Motion Sensors
- So on.

MOBILE WEB VS DESKTOP WEB

What is Mobile Web?

- The **mobile Web** refers to the use of browser-based Internet services from handheld **mobile** devices, such as smartphones or feature phones, through a **mobile** or other wireless network.

- From **User’s** Perspective - Mobile Web is basically just web content accessed from
a mobile device

- From **Developer’s** Perspective - It is a group of best practices, design patterns and new code to be learn in order to adapt the web to mobile devices.

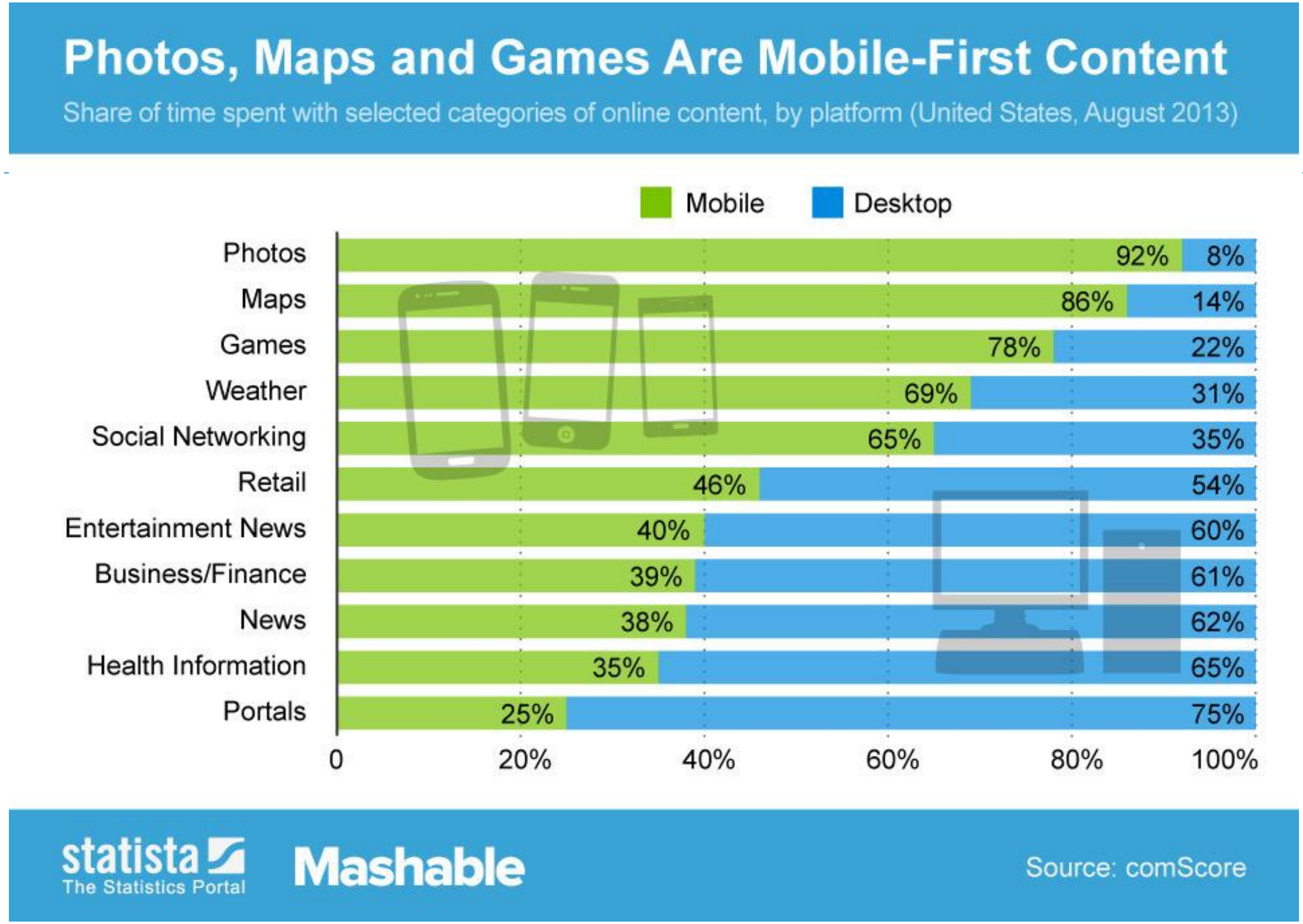
Mobile Web vs Desktop Web

Differences between Mobile Web and Desktop Web:

- Different browsing **experience**
- Different user **contexts**
- Different **user behavior** (no clear ‘pop-up’ contexts’)
- Radically different browsers DPI and screen aspect ratio
- **Contextual** rather than global navigation
- Platform **UX interactions**, tapable vs. Clickable
- Slower network and high latency (lag)
- Slower hardware and less memory

Mobile Web vs Desktop Web

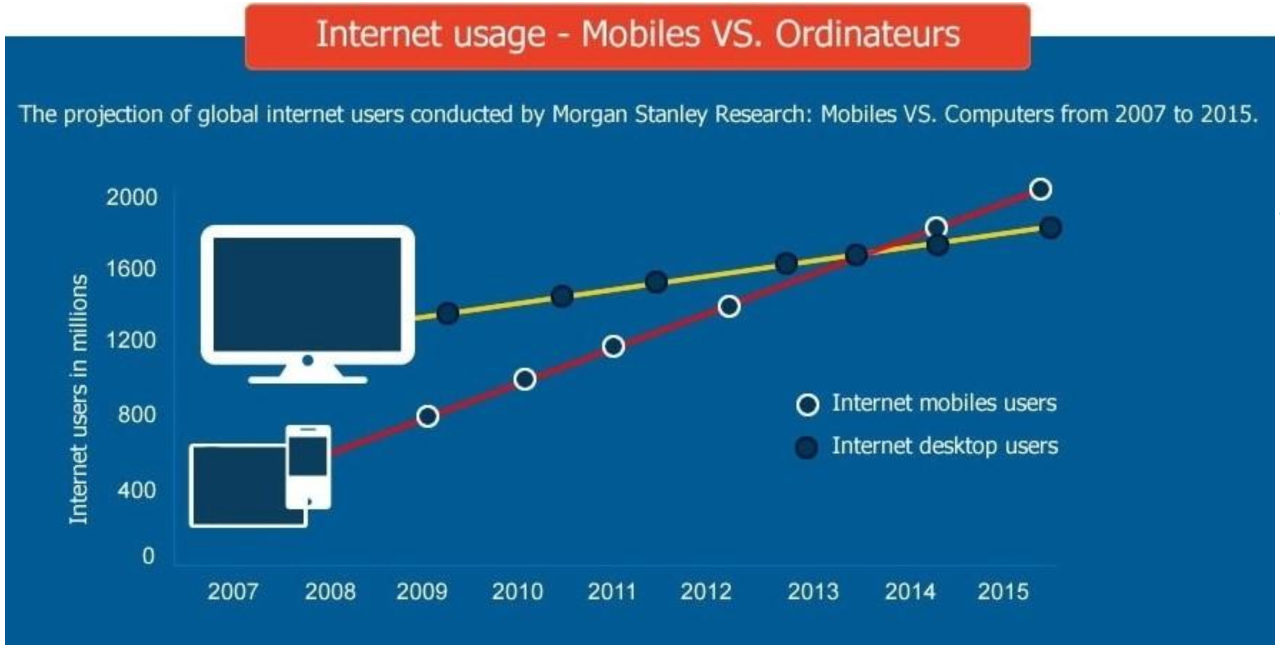
- When are you prefer using Desktop Web over Mobile Web, or vice versa?



https://www.statista.com/chart/1567/photos-maps-and-games-are-mobile-first-content/
https://martech.zone/mobile-marketing-statistics/
https://www.go-globe.com/digital-media-consumption-statistics-and-trends-infographic

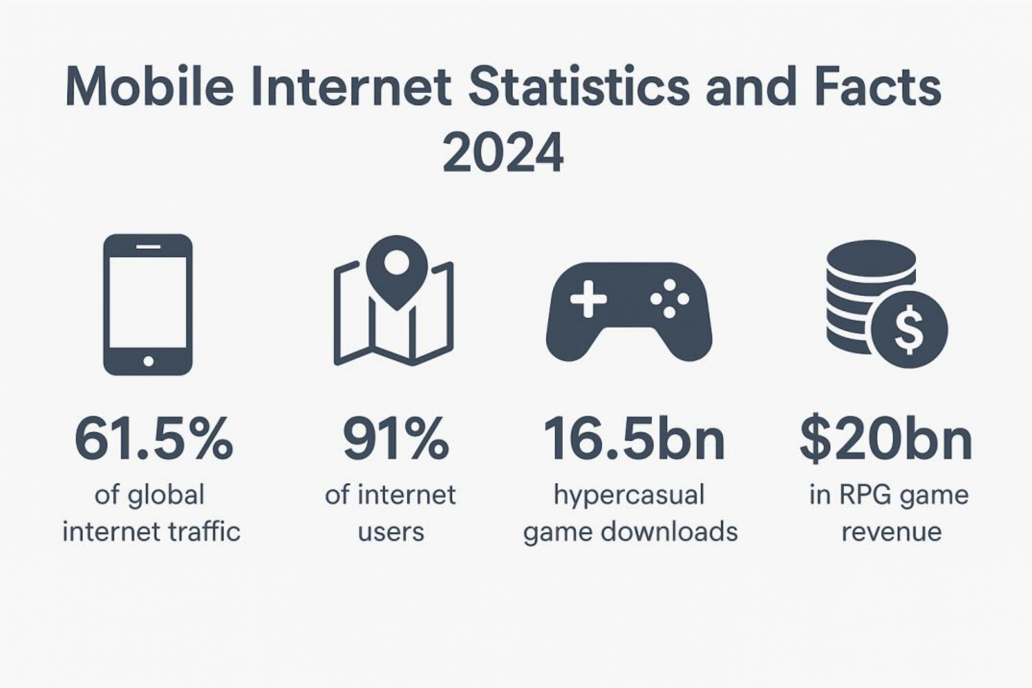
Why a Mobile Website is Important for Businesses Now:

- **Vast majority of users will abandon a non-mobile optimized website**.
- That means that if your website is not a mobile website… you **losing your potential customers** and sales, and losing to the competitors.
- More people are surfing the web for product and service information from their mobile devices than from their desktop computers than ever before.
- Businesses cannot ignore the mobile revolution any longer.

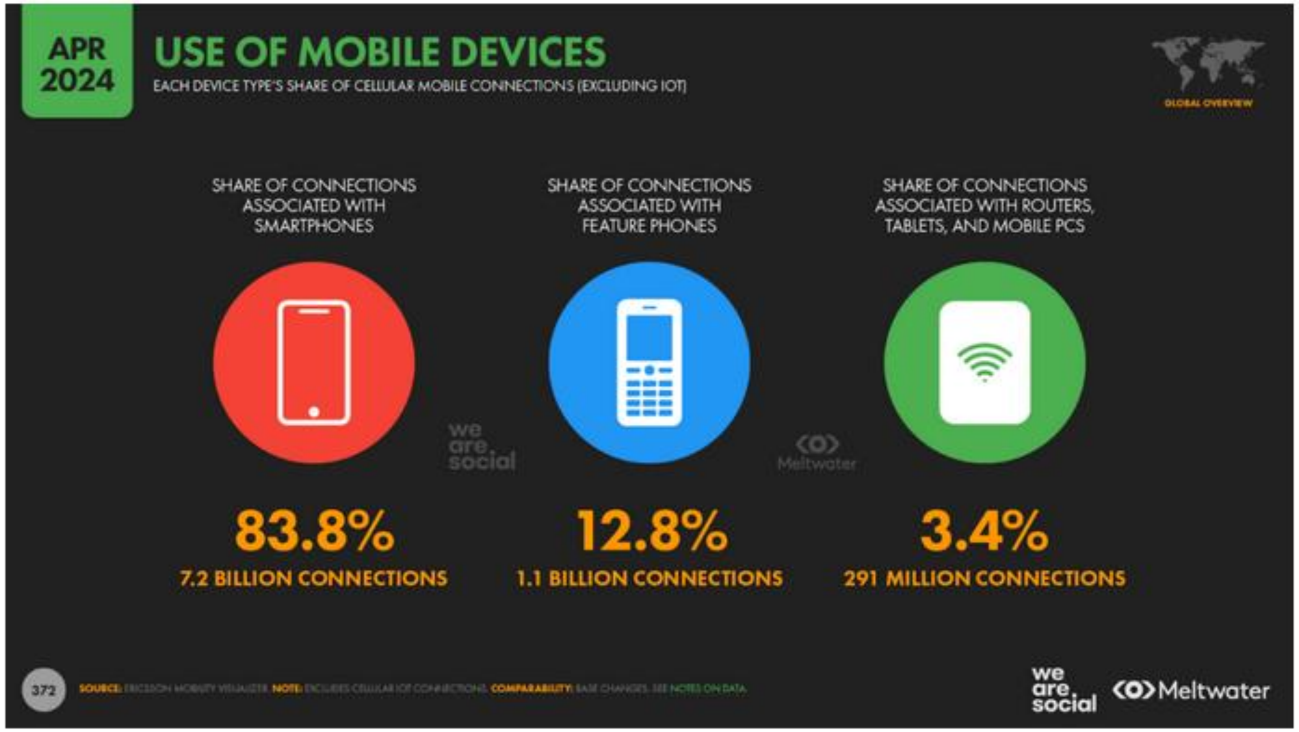


Internet usage - Mobiles VS. Ordinateurs
The projection of global internet users conducted by Morgan Stanley Research: Mobiles VS. Computers from 2007 to 2015.
Internet users in millions
2000
1600
1200
800
400
0
2007
2008
2009
2010
2011
2012
2013
2014
2015
Internet mobiles users
Internet desktop users

SMS



Mobile Internet Statistics and Facts 2024
61.5%
of global internet traffic
91%
of internet users
16.5bn
hypercasual game downloads
$20bn
in RPG game revenue



APR 2024
USE OF MOBILE DEVICES
EACH DEVICE TYPE'S SHARE OF CELLULAR MOBILE CONNECTIONS (EXCLUDING IOT)
SHARE OF CONNECTIONS ASSOCIATED WITH SMARTPHONES
SHARE OF CONNECTIONS ASSOCIATED WITH FEATURE PHONES
SHARE OF CONNECTIONS ASSOCIATED WITH ROUTERS, TABLETS, AND MOBILE PCS
83.8%
7.2 BILLION CONNECTIONS
12.8%
1.1 BILLION CONNECTIONS
3.4%
291 MILLION CONNECTIONS
372
we are social
Meltwater

MOBILE WEB ERA

Mobile Web Eras (1st Era: WAP 1.0)

Mobile Web has been divided into several Eras:

1. **WAP (Wireless Application Protocol) 1 Era**

   - Was the most popular protocol for accessing mobile web around the world except in Japan (they use i-mode)
   - Describes a protocol suite that allows the transformation between device and the internet via a WAP gateway. In WAP, Mobile Web are written in **WML**.
   - WAP is obsolete and most modern mobile phone browsers (particularly iPhone and Android) do not support WAP anymore.

Mobile Web Eras (2nd Era: WAP 2.0)

2. WAP 2.0 Era

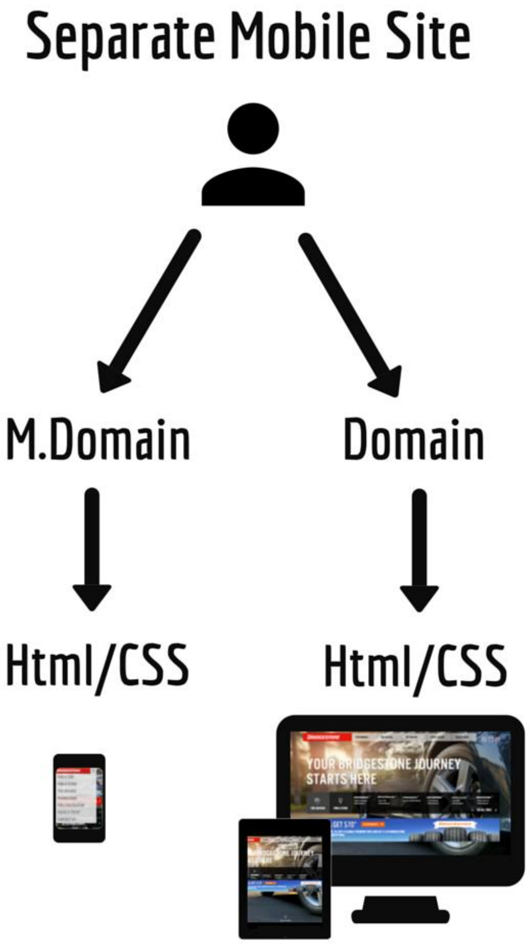
- It is a continuation from WAP 1 (1.0/1.1)
- WAP 2.0 deprecates WML and created **XHTML Mobile Profile**.
- Released in 2002 but become widely adopted in 2004
- WAP 2.0 standards is nearer to the web standards than the previous version and **allows HTTP communication** between device and the server
- The WAP 2.0 introduces the **'m' subdomain** for mobile web browsers

Mobile Web Eras (3rd Era: HTML5)

3. **HTML5 Era**

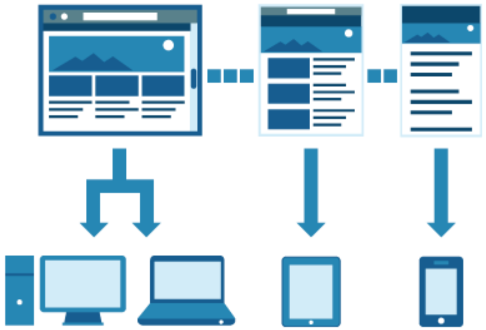
- The HTML 5 era devices starting to promote the idea of not having a different URL for the mobile and desktop web
- It promotes the concept of a **single URL** which can have different views for **mobile devices and desktop web**
- HTML5 era devices also promotes the concept of native-like ability to access device capabilities (such as accessing microphone, camera, graphic accelerators, GPS, etc)

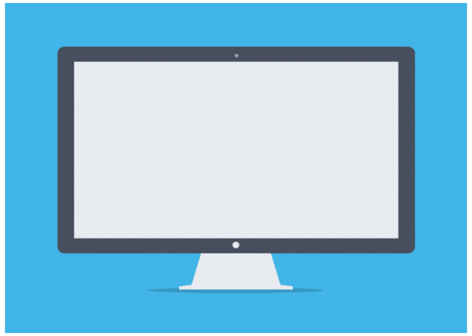
m Subdomain and Responsive Mobile

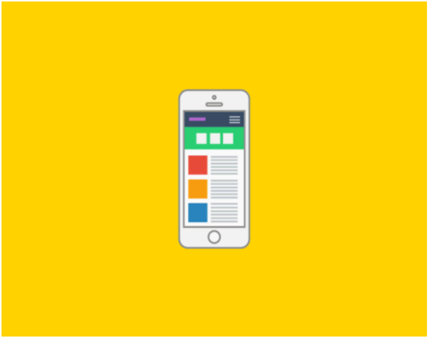


Responsive Web Design

- Creating flexible layouts that can adapt to suit the size or orientation of any device: **screen size, platform and orientation**







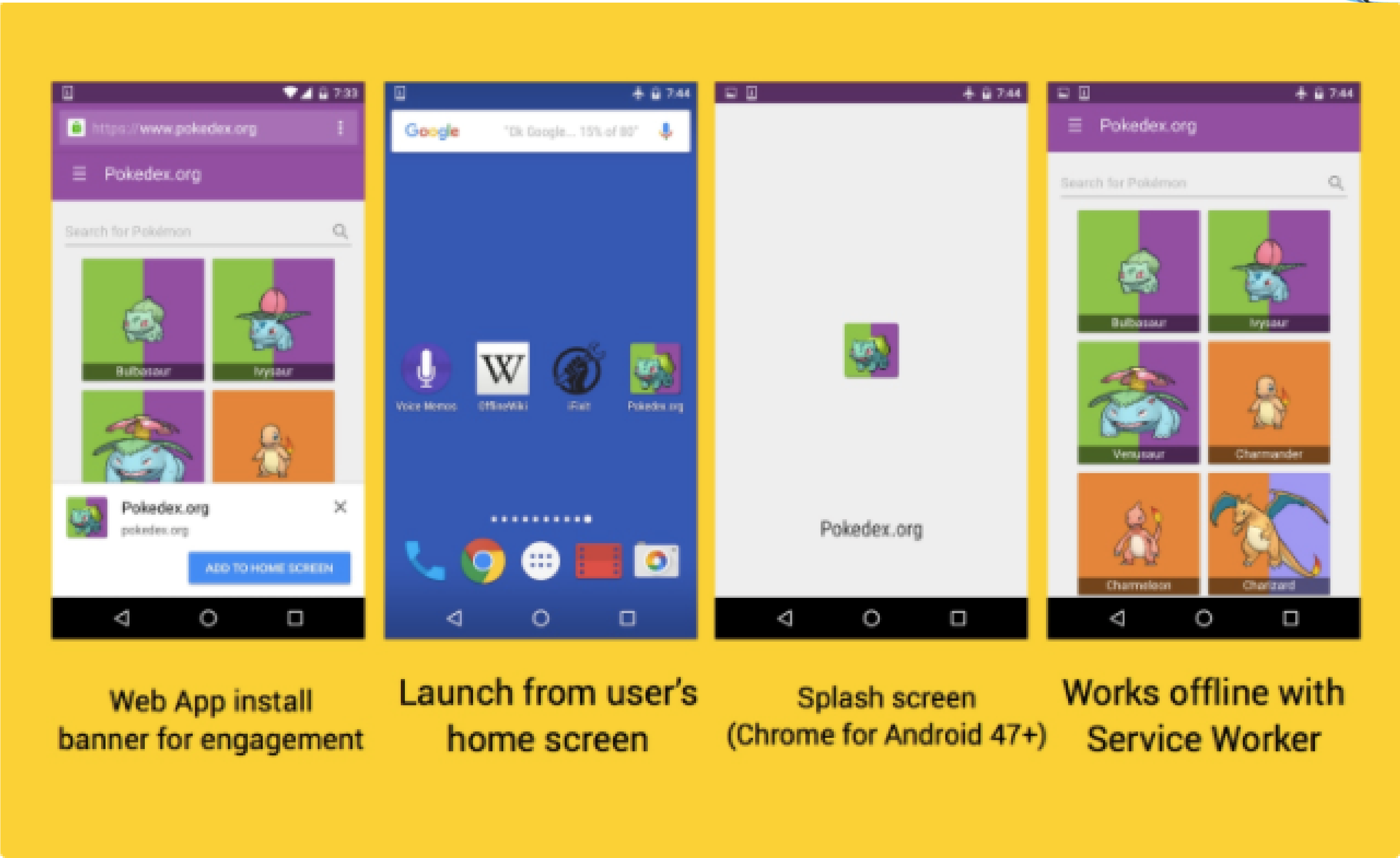
Advantages

- One **single HTML** document to be maintained
- One **single CSS** file to be maintained
- The site is **easily accessible** on any type of device.
- Better **user experience**.
- Responsive Web is **flexible and adaptable**

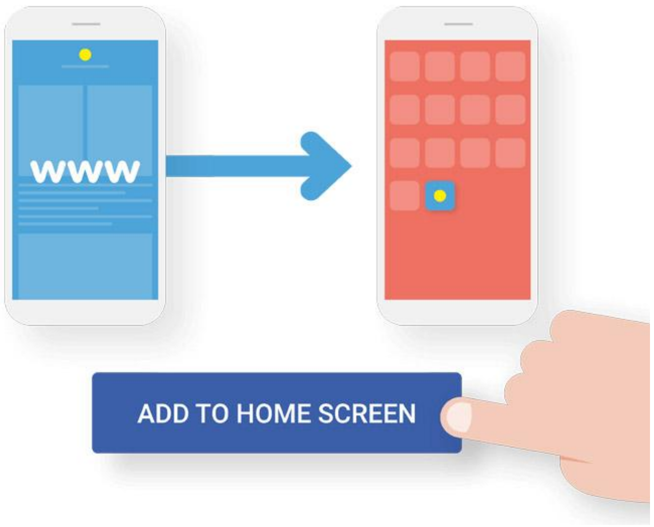
Progressive Web App

Progressive Web App

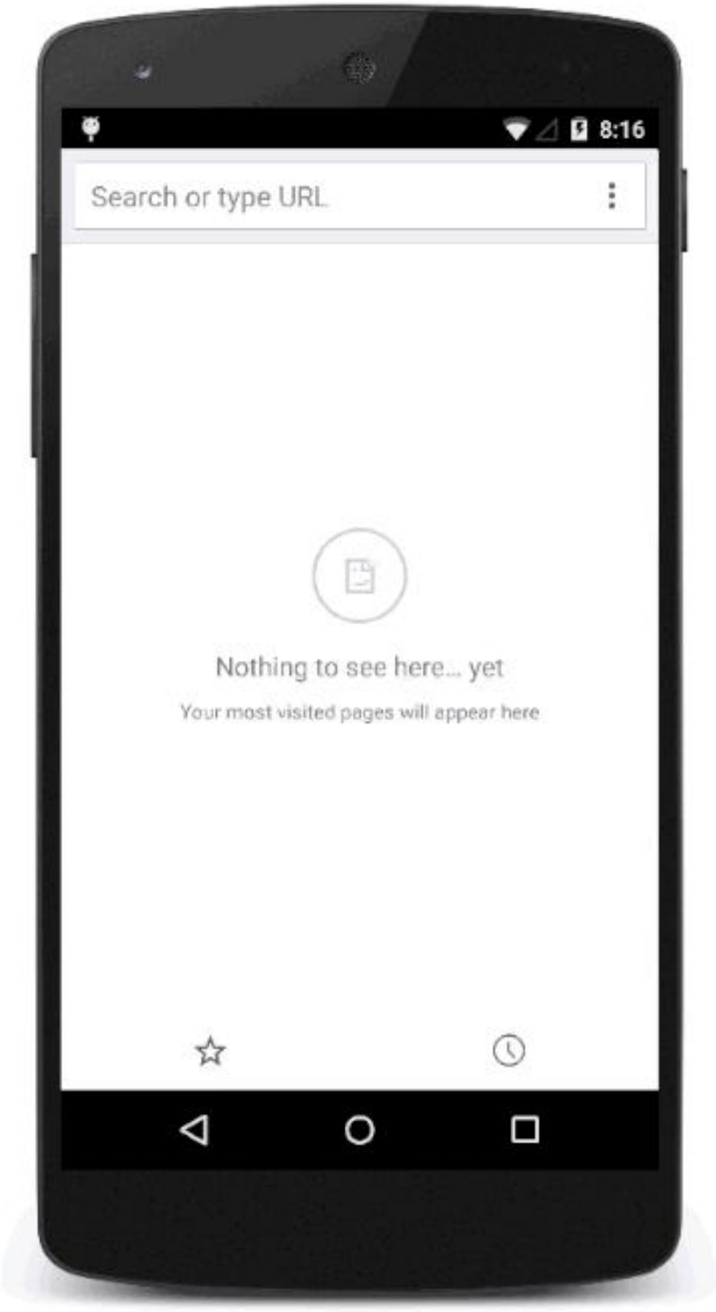
- web apps that load like regular websites but can offer the user functionality such as **working offline**, **push notifications**, and **device hardware access** traditionally available only to native mobile applications.
- Progressive web apps are not a new way to design web sites, but a way to make the web experience equal to native applications on mobile and desktop.
- https://www.youtube.com/watch?v=tdXGkthgNuk
- https://greenice.net/business-need-progressive-web-app/


Pokedex.org
ADD TO HOME SCREEN
Web App install banner for engagement
Launch from user's home screen
Splash screen (Chrome for Android 47+)
Works offline with Service Worker

Home Screen

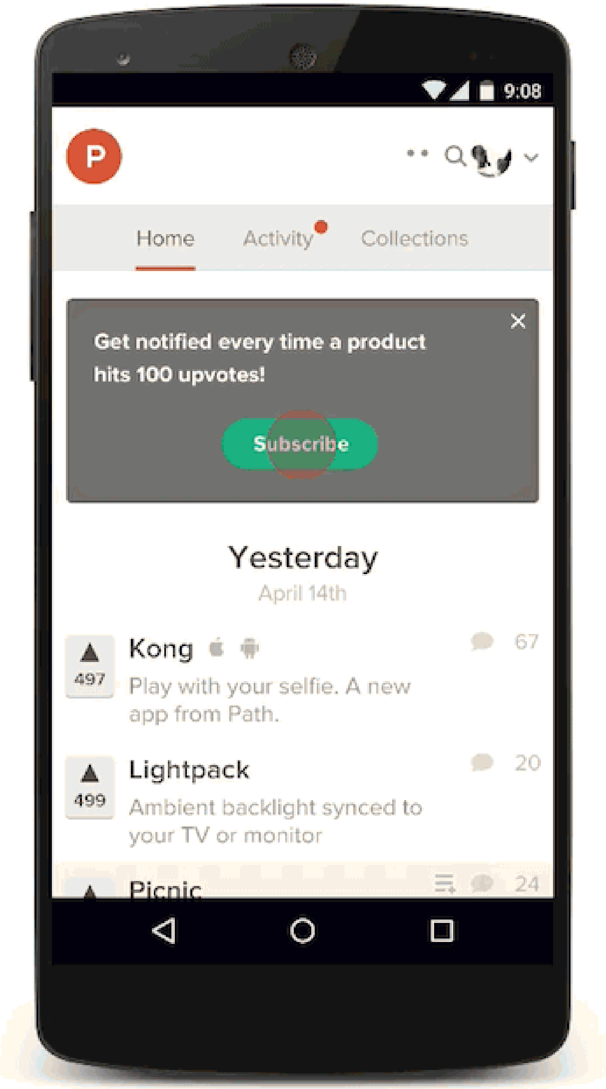


www
ADD TO HOME SCREEN

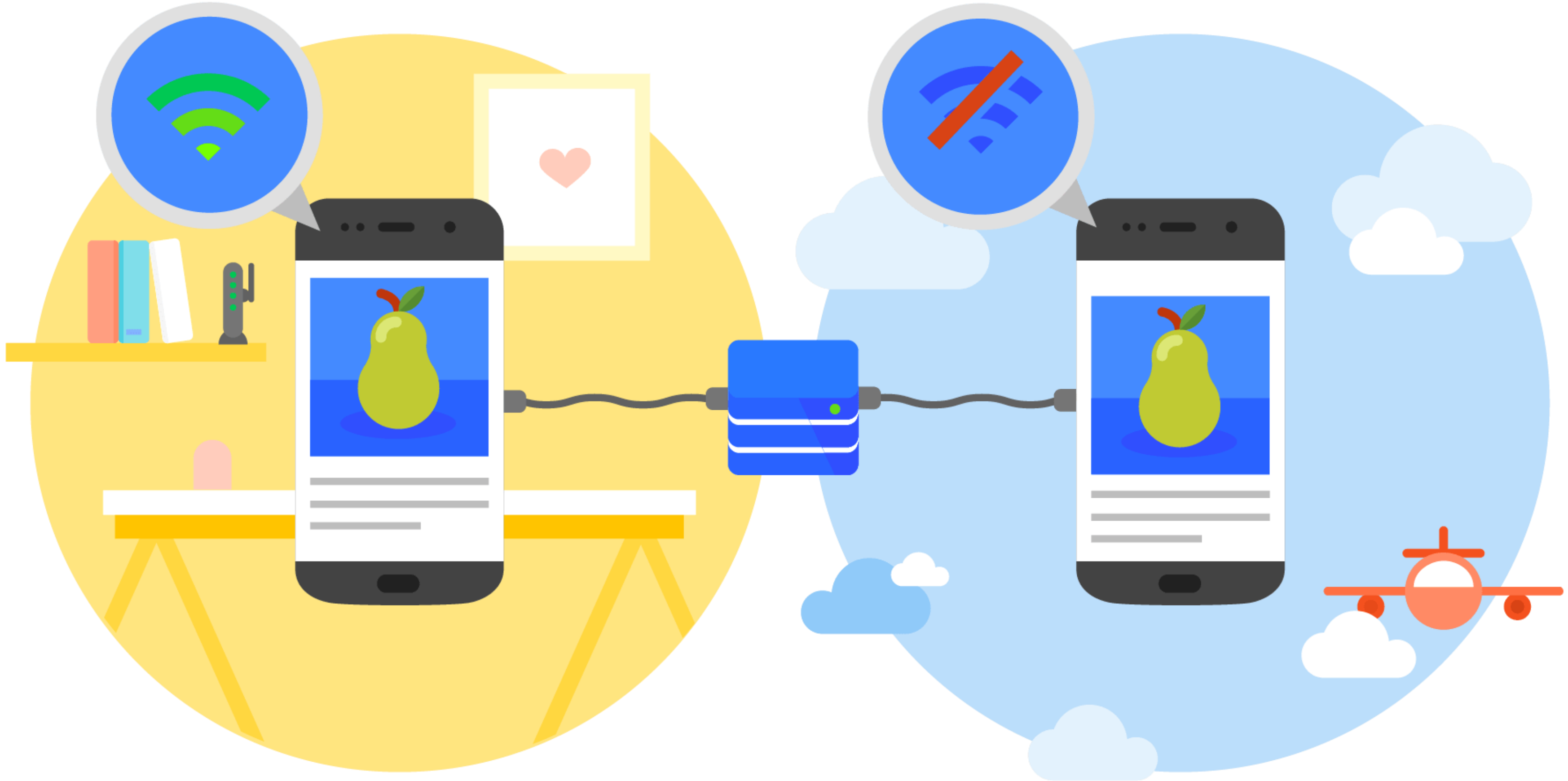


8:16
Search or type URL
Nothing to see here... yet
Your most visited pages will appear here

Push Notification



Offline Access



THE MOBILE BROWSING EXPERIENCE

The Mobile Browsing Experience

- A mobile website can be navigated using different techniques.
- Every mobile browser uses one or many of these modes of navigation.

1. **Cursor Navigation**
2. **Focus Navigation**
3. **Touch Navigation**
4. **Multitouch Navigation**

Cursor Navigation

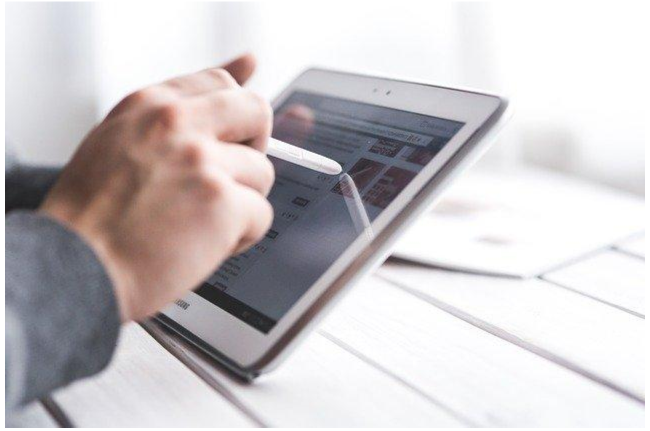
- **Emulates a mouse cursor** over the screen that can be moved using arrow keys.
- In this navigation paradigm, a **mouse click is emulated with the Fire or Enter key**
- Cursor Navigation is used in devices which supports
  - "hardware keypad"
  - "hardware navigational d-pad"

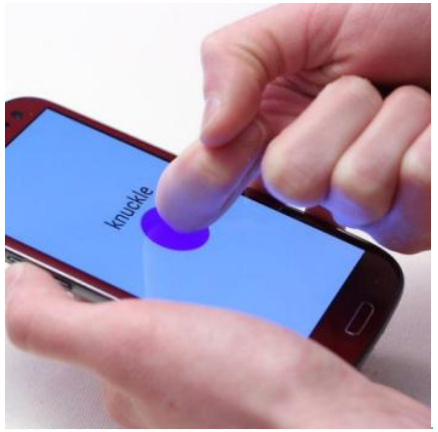
Focus Navigation

- In this mode, **a border or a background color is used to show where the focus is**
- Most frequent navigation used for browsing web on low-end and mid-range devices.
- used for non-touch devices
- The user **uses the cursor keypad** to navigate between links and scroll the website

Touch Navigation

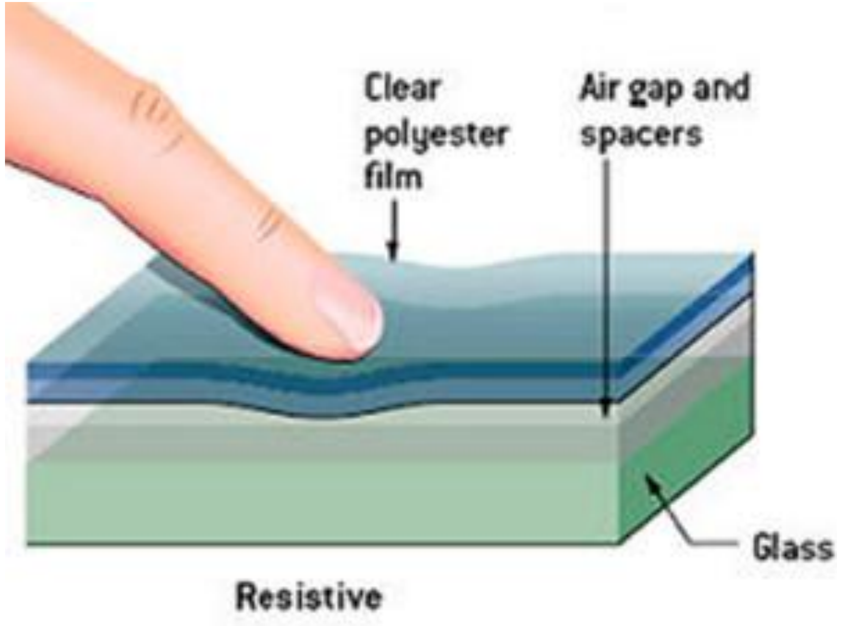
- In touch navigation, the user may **navigate using a finger or a stylus.**
- Touch devices allow user to use detectable gesture to easily perform some actions.
- Touch navigation is available to devices with **Resistive** and **Capacitive touch screens**





Resistive Touch Screen

- **Uses pressure** - the pressure you apply causes the screen to respond.
- You **can use anything** you want on a resistive touchscreen to make the touch interface work
  - a gloved finger,
  - a wooden rod,
  - a fingernail



Resistive Touch Screen

Resistive Touch Screen

Advantages

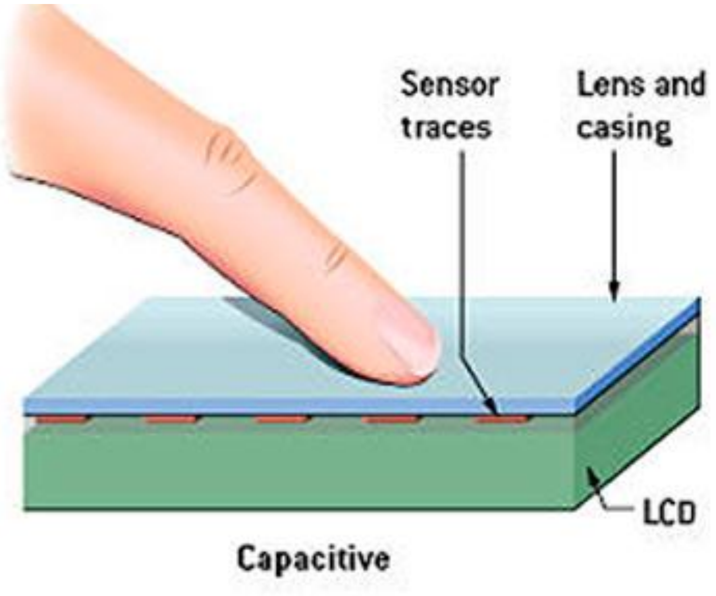
- More accurate, useful for handwriting recognition
- can also be operated with any pointing device
- Cheaper
- Can be operated even with a Gloved Fingers

Disadvantages

- Visibility under sunlight is poor
- Multi-touch is not possible
- vulnerable to scratches

Capacitive touch screens

- Makes **use of the electrical** properties of the human body.
- Highly accurate and respond instantly when touched by a human finger.
- Rely on the electrical change caused by a light touch of a finger. This is the reason you cannot use a capacitive screen while wearing gloves



Capacitive touch screens

Capasitive Touch Screen

Advantages

- Multi-touch is possible
- Visibility is good even in sunlight
- Glass can be used as the outer layer.
- More responsive

Disadvantages

- Cannot use standard stylus
- Finger size constraint and limitation
- Expensive

Glove for Capacitive touch screens

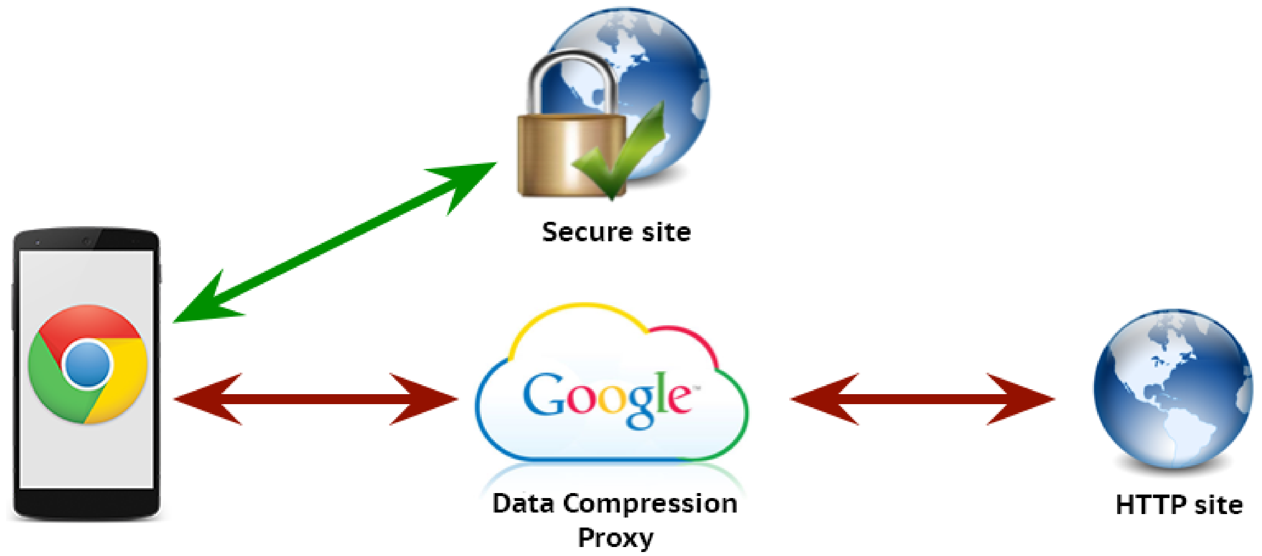
DIRECT VS CLOUD-BASED / PROXIED BROWSERS

Direct Browsers

- Direct browsers are type of mobile web browsers that **connect directly to the web site**
  - Get content directly from the website server
- Mobile web site is **accessed and formatted on the mobile devices** and displayed to the users
- Direct browsers are preferred by users who values their privacy as it does not perform connection via proxy

Cloud-based (or Proxied) browsers

- Cloud based or Proxied browsers is a relatively newer concept of mobile browsers.
- Cloud browsers serves the mobile pages to users through a **proxy on the cloud**



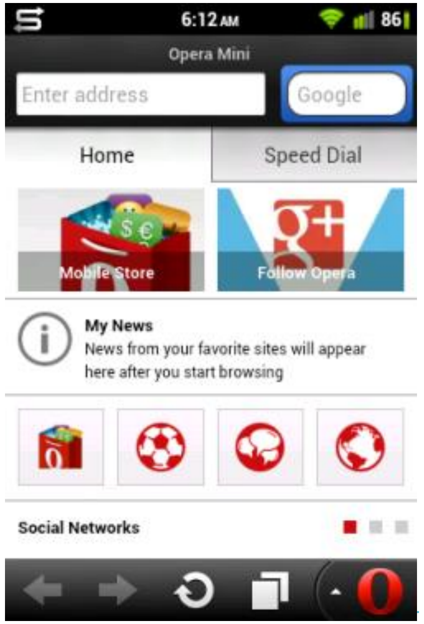
Cloud-based (or Proxied) browsers

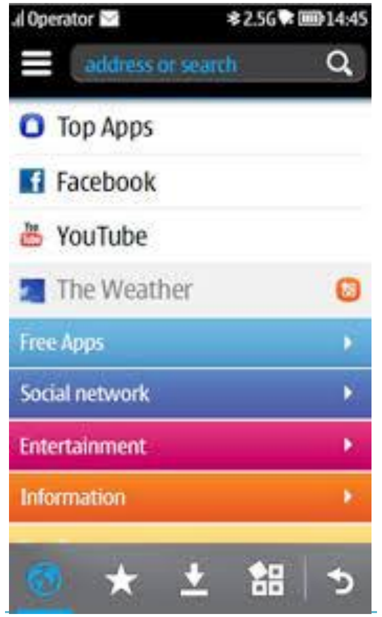
- The proxy is usually used to perform several optimization on the fly to benefit mobile users, among them:

1. **Reduces** content, eliminating features that are not mobile-compatible
2. **Compressed** the content (including images)
3. **Pre-renders** the content, so it can be displayed in the browser faster
4. **Convert** content (converts flash video files, convert images file into format that can be viewed by mobile browsers
5. **Encrypts** the content
6. **Caches** content for quick access for a later time

Cloud-based (or Proxied) browsers

- Cloud-based browsers **consume less bandwidth**, **increase performance** and **reduced total navigation time**.
- Example :Opera Mini, Nokia Xpress Browser and Google Chrome (if ‘optimize for mobile’ option activated





Drawbacks of Cloud/Proxied Browsers

- Some users do not prefer their connection to go through proxy due to **privacy concerns**
- Multimedia or images **quality may be degraded** due to compression
- May lead to **inconsistent page representation**
- For developers – cloud browsers may render the page differently from direct browsers

END